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(12) **United States Design Patent** (10) **Patent No.:** **US D1,090,782 S**
Blum (45) **Date of Patent:** **** Aug. 26, 2025**

(54) **FLOW REGULATOR**

(56) **References Cited**

(71) Applicant: **Neoperl GmbH**, Mullheim (DE)

U.S. PATENT DOCUMENTS

(72) Inventor: **Gerhard Blum**, Gutach (DE)

6,126,093 A 10/2000 Grether
D536,417 S 2/2007 Buttgen
(Continued)

(73) Assignee: **Neoperl GmbH**, Müllheim (DE)

OTHER PUBLICATIONS

(**) Term: **15 Years**

(21) Appl. No.: **29/825,755**

12 Pcs Faucet Aerator for Cache Aerators, Nov. 10, 2020, amazon.com, Oct. 21, 2021.URL: <https://www.amazon.com/dp/B08N4G3XJ8> (Year: 2020).

(22) Filed: **Feb. 4, 2022**

(Continued)

Related U.S. Application Data

(62) Division of application No. 29/685,882, filed on Apr. 1, 2019, now Pat. No. Des. 946,110, which is a division of application No. 29/598,490, filed on Mar. 27, 2017, now Pat. No. Des. 849,887, which is a division of application No. 29/533,780, filed on Jul. 22, 2015, now Pat. No. Des. 786,400, which is a division of application No. 29/460,497, filed on Jul.

(Continued)

Primary Examiner — Keith J Wilson

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(57) **CLAIM**

The ornamental design for a flow regulator as shown and described.

DESCRIPTION

FIG. 1 is a top plan view of a first embodiment of a flow regulator in accordance with my design;
FIG. 2 is a top, front, right perspective view thereof;
FIG. 3 is a bottom plan view thereof;
FIG. 4 is a bottom, front, right perspective view thereof;
FIG. 5 is a right side elevational view thereof, with the left side elevational view being a mirror image thereof; and
FIG. 6 is a cross-sectional view taken along line 6-6 in FIG. 3;
FIG. 7 is a top plan view of a second embodiment of a flow regulator in accordance with my design;
FIG. 8 is a top, front, right perspective view thereof;
FIG. 9 is a bottom view thereof;
FIG. 10 is a bottom, front, right perspective view thereof;
FIG. 11 is a right side elevational view thereof; with the left side elevational view being a mirror image thereof; and,
FIG. 12 is a cross-sectional view taken along line 12-12 in FIG. 9.

The broken lines represent portions of the flow regulator that form no part of the claimed design.

(30) **Foreign Application Priority Data**

Dec. 22, 2010 (EP) 001251946-01/163

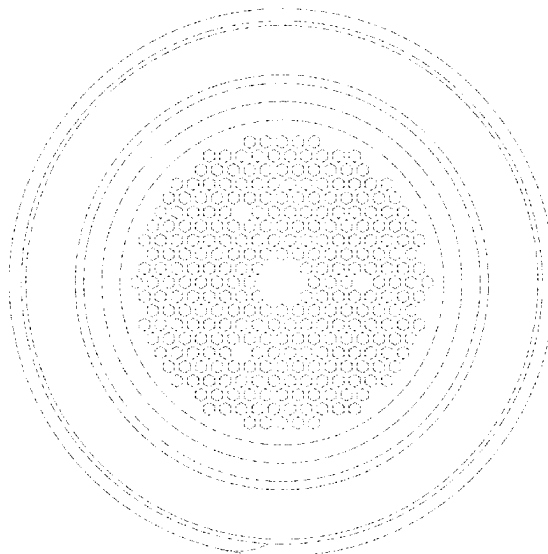
(51) **LOC (15) Cl.** **23-01**

(52) **U.S. Cl.**
USPC **D23/213**

(58) **Field of Classification Search**
USPC D23/207, 209, 213, 214, 215, 223, 224,
D23/225, 226, 229, 230, 249
CPC A61H 9/0021; A61H 33/00; B05B 1/00;
B05B 1/14; B05B 1/185; B05B 1/08;
B05B 1/02; B05B 1/26; B05B 12/002;
B05B 1/18; B05B 9/01

See application file for complete search history.

1 Claim, 6 Drawing Sheets



Related U.S. Application Data

11, 2013, now Pat. No. Des. 737,403, which is a division of application No. 29/388,860, filed on Apr. 4, 2011, now Pat. No. Des. 690,796.

(56)

References Cited

U.S. PATENT DOCUMENTS

D537,916 S 3/2007 Buttgen
D539,391 S 3/2007 Grether
D550,816 S 9/2007 Grether
D550,817 S 9/2007 Grether
D550,818 S 9/2007 Grether
7,357,337 B2 4/2008 Ferrari
D577,416 S 9/2008 Buttgen
D591,403 S 4/2009 Grether
D605,732 S 12/2009 Stein
D612,911 S 3/2010 Grether
7,731,107 B2 6/2010 Grether
D652,896 S 1/2012 Grether
RE43,347 E 5/2012 Grether
8,205,810 B2 6/2012 Lacher
8,328,119 B2 12/2012 Grether
D688,774 S 8/2013 Tempel
D690,796 S 10/2013 Blum
D690,797 S * 10/2013 Blum D23/249
D728,738 S 5/2015 Birmelin
D728,740 S 5/2015 Tempel
D733,255 S * 6/2015 Blum D23/213
D737,403 S 8/2015 Blum
D752,710 S 3/2016 Tempel
D753,792 S * 4/2016 Blum D23/213
D754,282 S * 4/2016 Blum D23/249
9,416,522 B2 8/2016 Tempel
D776,240 S 1/2017 Haung
D776,242 S * 1/2017 Blum D23/213
9,580,893 B2 2/2017 Blum
D786,400 S * 5/2017 Blum D23/213
D788,264 S 5/2017 Stein
D788,881 S 6/2017 Birmelin
D791,921 S * 7/2017 Blum D23/249
D800,250 S * 10/2017 Blum D23/213
D800,867 S * 10/2017 Blum D23/249
D812,722 S * 3/2018 Blum D23/249
D828,492 S 9/2018 Stein et al.
D833,573 S * 11/2018 Blum D23/213
D833,583 S 11/2018 Birmelin
D834,153 S 11/2018 Tempel
D839,385 S * 1/2019 Vögeli D23/213
D840,503 S * 2/2019 Stein D23/213
D844,104 S * 3/2019 Birmelin D23/213
D844,747 S 4/2019 Blum

D849,887 S * 5/2019 Blum D23/213
D849,888 S * 5/2019 Schürle D23/213
D851,208 S * 6/2019 Birmelin D23/213
D853,524 S * 7/2019 Song D23/213
D855,756 S * 8/2019 Blum D23/213
D860,402 S * 9/2019 Blum D23/249
D866,709 S 11/2019 Schurle
D866,710 S * 11/2019 Blum D23/213
D871,553 S * 12/2019 Blum B29C 49/60
D894,338 S * 8/2020 Lai D23/249
D895,065 S * 9/2020 Lai E03C 1/08
D896,343 S * 9/2020 Stein D23/213
D905,199 S * 12/2020 Wu D23/213
D905,820 S * 12/2020 Weiß E03C 1/08
D906,478 S * 12/2020 Stein D23/213
D911,488 S * 2/2021 Spater E03C 1/08
D921,152 S * 6/2021 Blum D23/213
D926,923 S * 8/2021 Ross D23/249
11,097,290 B2 8/2021 Wu D23/213
D933,162 S 10/2021 Stein E03C 1/084
D933,776 S * 10/2021 Blum D23/213
D936,787 S * 11/2021 Morgan E03C 1/084
D937,383 S * 11/2021 Thibodeau D23/213
D937,384 S * 11/2021 Khubani B29C 49/60
D946,110 S * 3/2022 Blum D23/213
D946,707 S * 3/2022 Blum E03C 1/084
D1,001,964 S * 10/2023 Stein D23/249
2011/0006132 A1 1/2011 Grether D23/213
2011/0266369 A1 11/2011 Grether
2016/0273199 A1 * 9/2016 Blum E03C 1/08
2019/0301148 A1 * 10/2019 Schurle E03C 1/08
2020/0199857 A1 * 6/2020 Stein E03C 1/086
2020/0224395 A1 7/2020 Stein
2021/0277639 A1 * 9/2021 Birmelin E03C 1/086
2021/0285194 A1 * 9/2021 Stein E03C 1/084
2021/0388586 A1 * 12/2021 Stein E03C 1/084

OTHER PUBLICATIONS

JQK Faucet Aerator, 2.2 GPM Flow Retrictor Insert Faucet Aera-tors, Dec. 13, 2017, amazon.com, Oct. 21, 2021 <https://www.amazon.com/JQK-Retrictor-Aerators-Replacement-Bathroomn/dp/B0774JJJD> (Year: 2019).

* cited by examiner

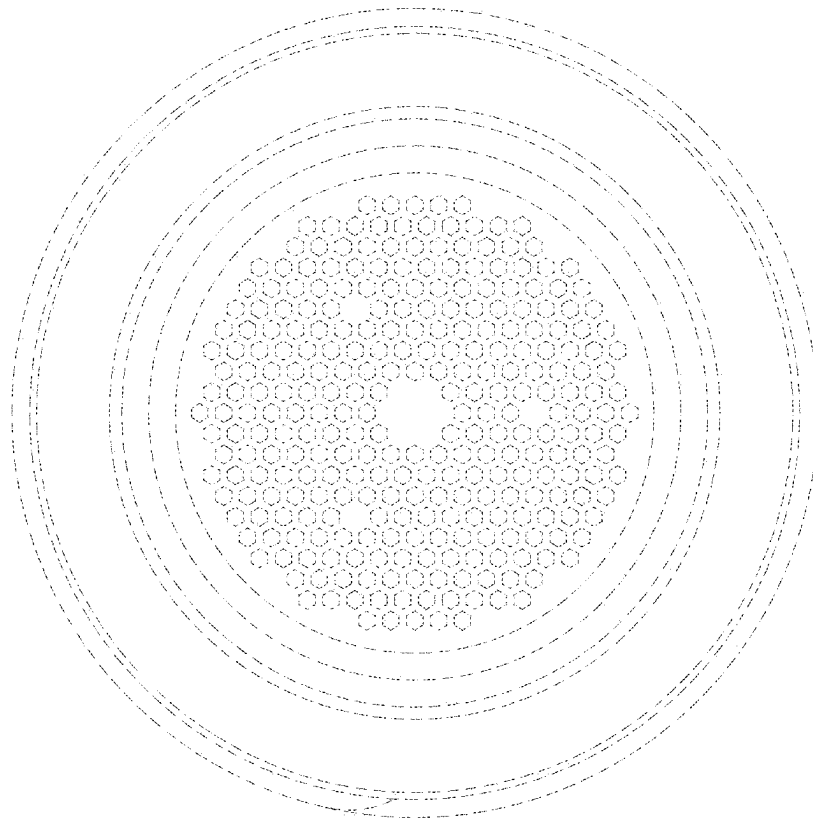


Fig. 1

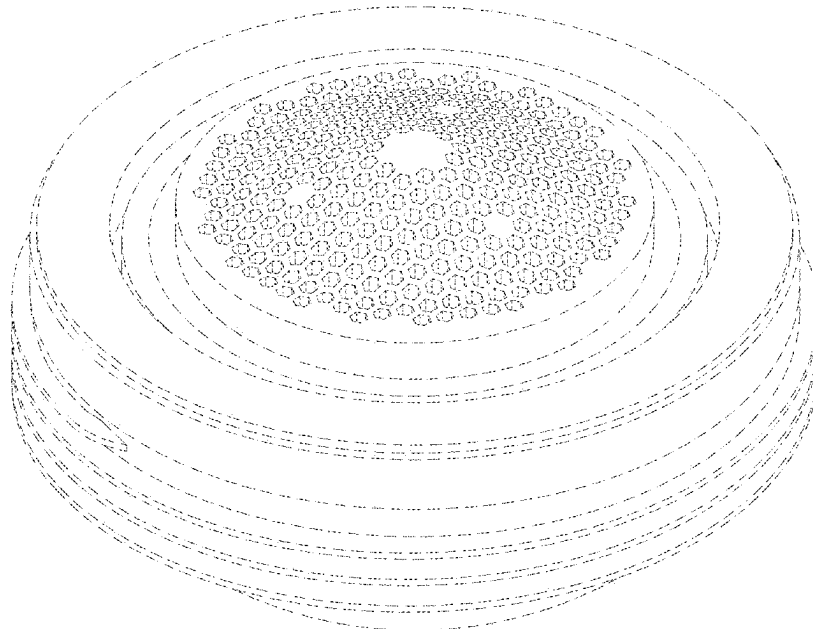


Fig. 2

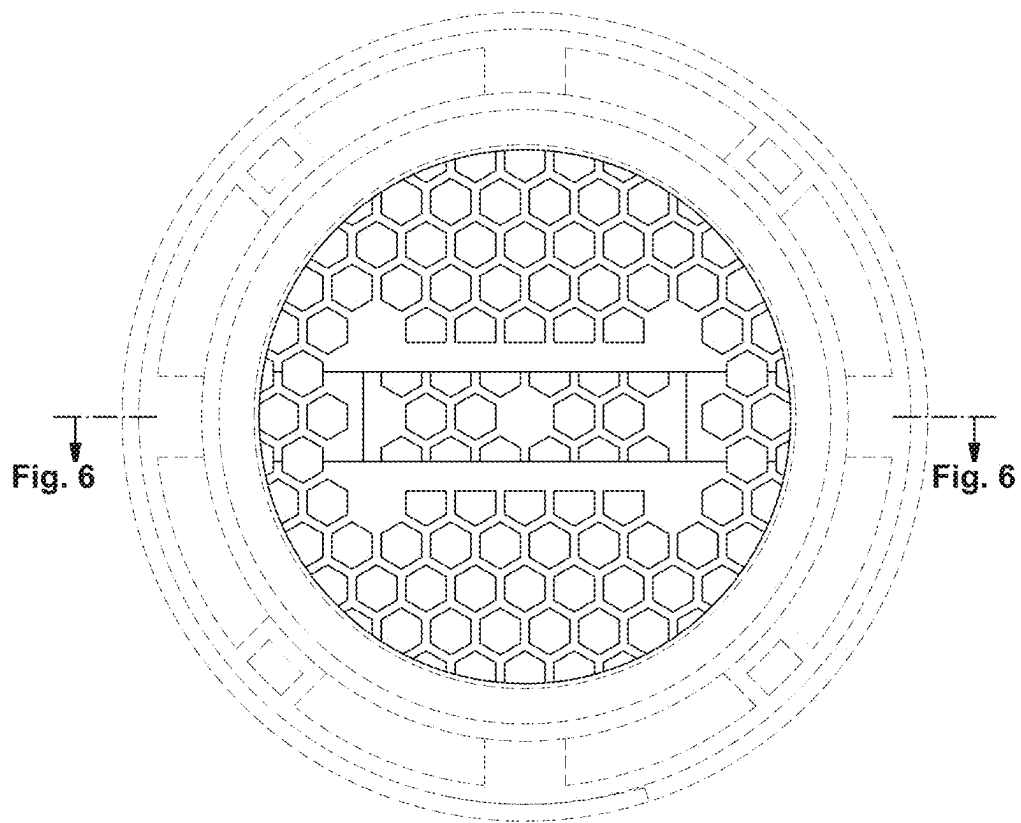


Fig. 3

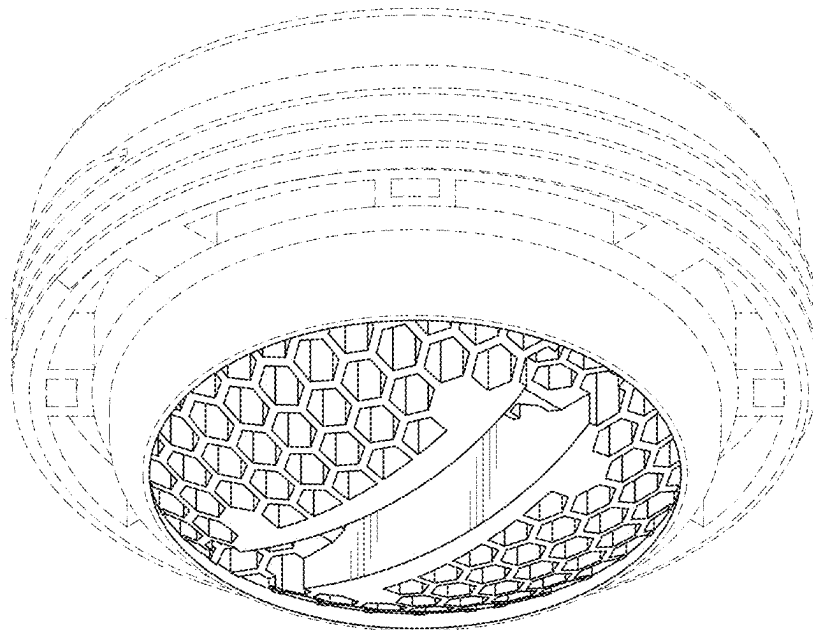


Fig. 4

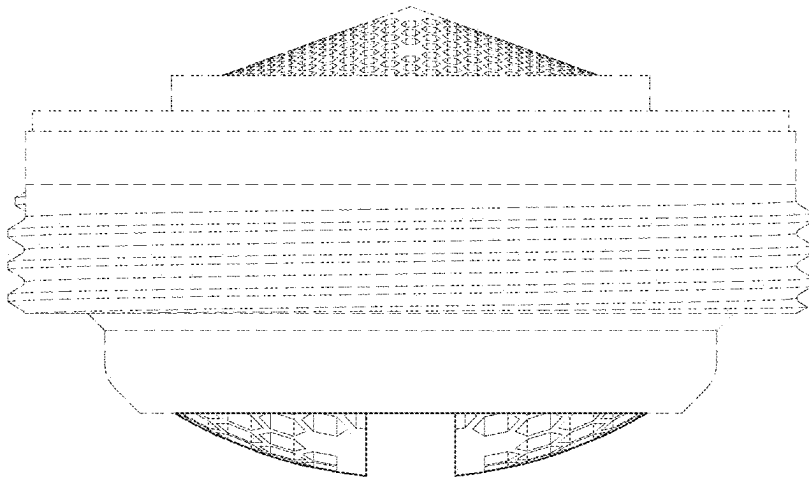


Fig. 5

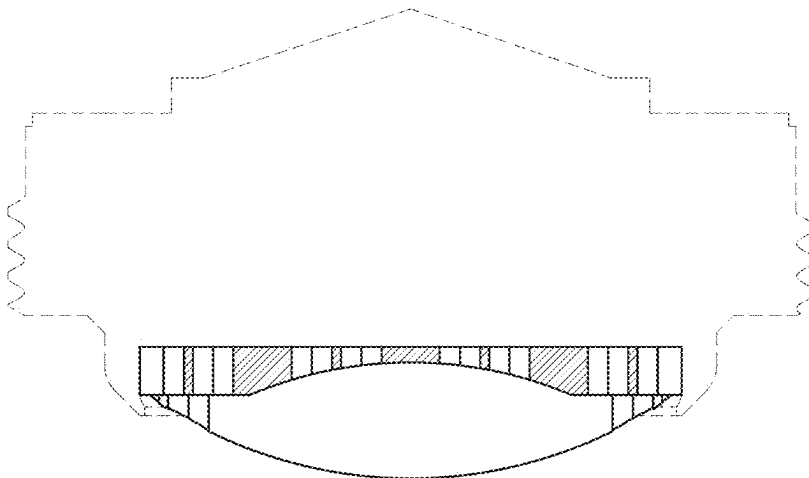


Fig. 6

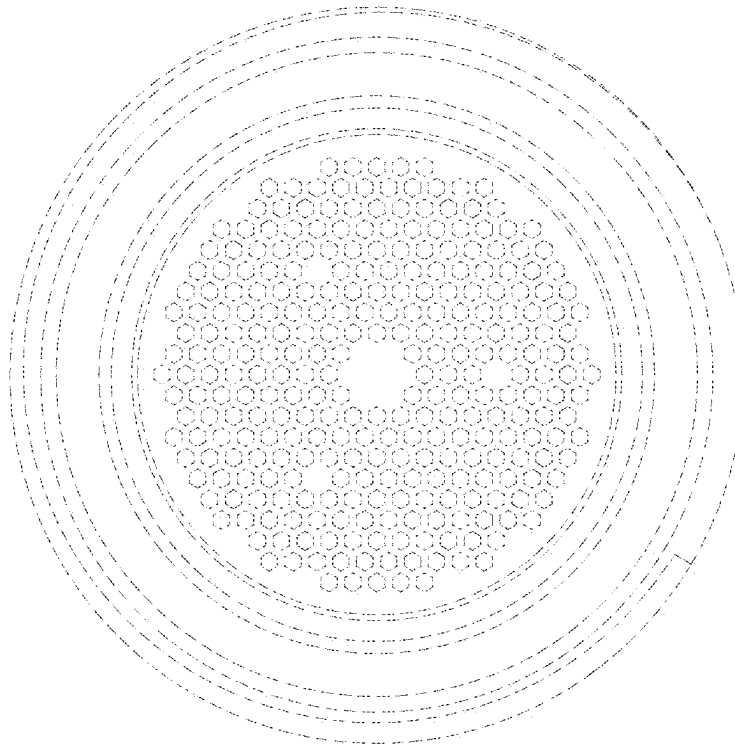


Fig. 7

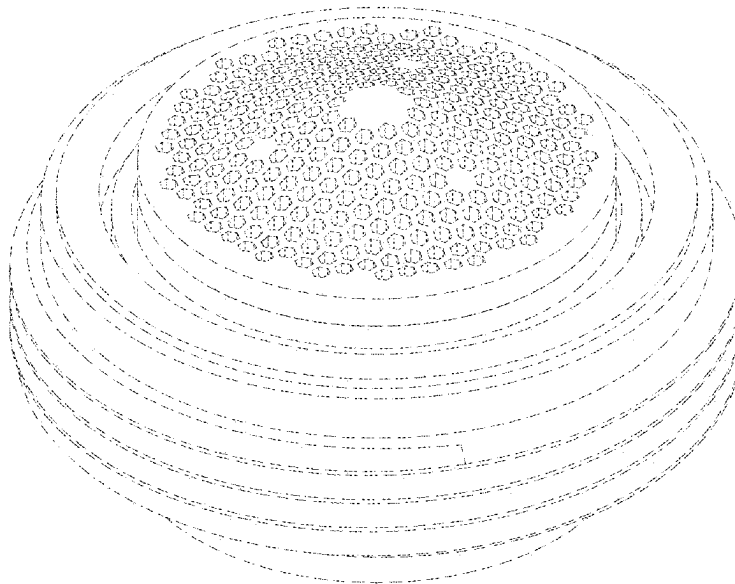


Fig. 8

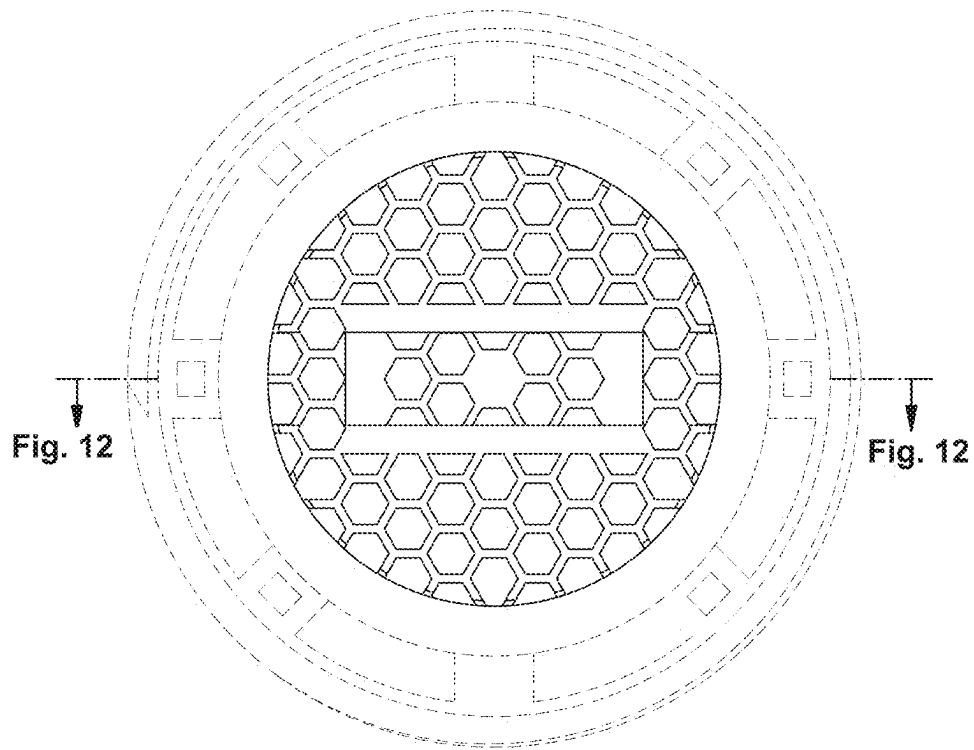


Fig. 9

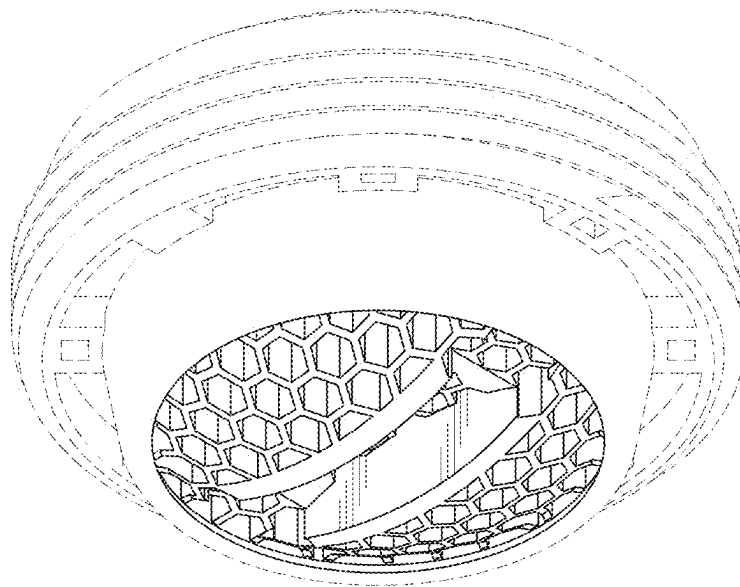


Fig. 10

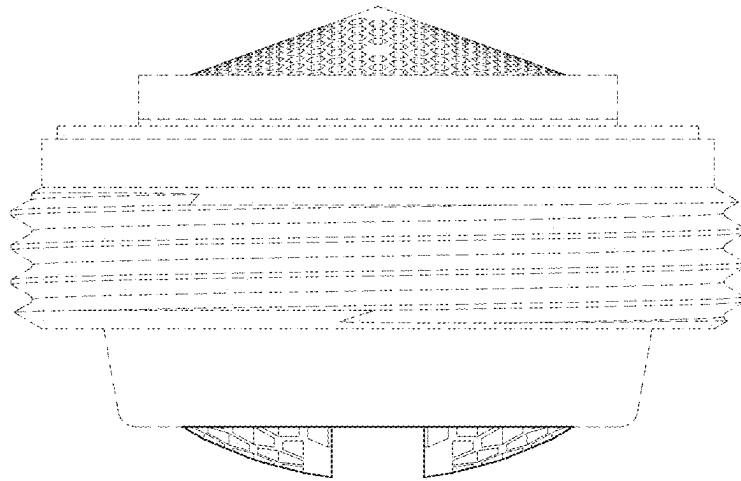


Fig. 11

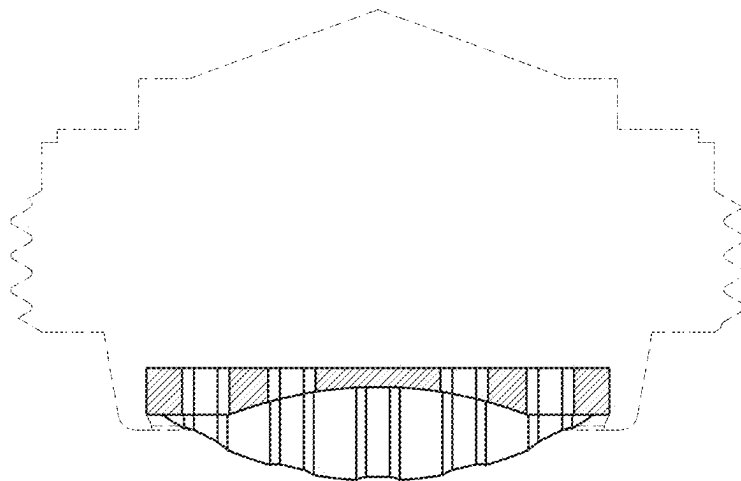


Fig. 12